

Patient-Reported Outcomes

Validity and Reliability of the EQ-HWB in Parents of Children With Health Conditions

Cate Bailey, PhD, Kim Dalziel, PhD, Leanne Constable, MPH, Tessa Peasgood, PhD

ABSTRACT

Objectives: The EQ Health and Wellbeing (EQ-HWB) is a new, generic instrument of health and wellbeing for patients and caregivers. It includes 2 measures, the EQ-HWB (25 items) and the EQ-HWB-S (9 items). Validating the instrument in a range of caregiver populations is now required. We investigated the validity and reliability of the EQ-HWB measures for caregivers, specifically for known-group differences in caregiver contribution, support, finance, work, and health.

Methods: The sample consisted of parents of children with chronic health conditions. We explored EQ-HWB item response distributions. Convergent validity of the EQ-HWB was assessed against the EQ-5D and CarerQol-7D using Spearman correlations. Known-groups validity analysis was conducted on caregiver outcomes using student's *t* tests and one-way analysis of variances for EQ-HWB and EQ-HWB-S sum scores, and EQ-HWB-S, EQ-5D-5L, and CarerQol index scores. Test-retest reliability was conducted over 2 days using intraclass correlation coefficients for totals and Kappa for items.

Results: The physical EQ-HWB items (seeing, hearing, mobility, activities, personal-care, and felt unsafe) were more negatively skewed than other EQ-HWB items. Known-group validity results conformed to a priori hypotheses; effect size comparisons between instruments varied. A priori hypotheses for convergent validity tests were all supported between EQ-HWB and EQ-5D-5L and most were supported between EQ-HWB and CarerQol. intraclass correlation coefficients results were good; Kappa results suggested moderate test-retest agreement, with variation across items.

Conclusions: The EQ-HWB instruments appeared valid and reliable against our criteria, and distinguished between groups for caregiver contribution, support, finance/income, and caregiver health. The EQ-HWB instruments seem suitable for measuring the quality of life of caregivers of children with health conditions.

Keywords: Caregiving, EQ-HWB, psychometrics, reliability, validity.

VALUE HEALTH. 2025; 28(9):1370–1379

Highlights

- The EQ Health and Wellbeing (EQ-HWB) is a new, generic instrument developed to measure quality of life for economic analysis across a range of health, social care and caregiver settings. Previous studies have investigated the validity of the instrument in caregivers of people living with dementia, children whose families have experienced adverse events, and children with health conditions. It is important to validate the instrument in a range of settings, including both patients and caregiver samples.
- We investigated the validity and reliability of the EQ-HWB in caregivers of children with health conditions in terms of response distribution, convergent validity, known-groups validity, and test-retest reliability analyses. Both EQ-HWB instruments performed well in psychometric tests, performed better than the EQ-5D on caregiver specific variables, and better than the CarerQol overall.
- Further studies are required to investigate the use of the EQ-HWB instruments in a range of populations; however, our study suggests that the EQ-HWB is valid and reliable for use in parents of children with health conditions. Specifically, our study showed that the EQ-HWB could differentiate between groups on caregiver contribution and the impact on caregiver's finance and ability to work from caregiving.

Introduction

The EQ Health and Wellbeing (EQ-HWB) instrument aims to measure broad aspects of quality of life that are important to both patients and caregivers, to allow for the calculation of quality-adjusted life-years (QALYs) for economic evaluation (developed as part of the “Extending the QALY” initiative).^{1–3} Widely used health-related quality-of-life instruments, such as the EQ-5D⁴ or the SF-6D⁵ are well suited to the health sector but may not capture important aspects of health and well-being in social care settings and caregiver populations.^{6–8} The EQ-HWB may fill this gap. Currently the EQ-HWB holds experimental status and is only available for use by researchers for validation purposes; future status updates will be announced on the EuroQol EQ-HWB website.⁹ Investigating the validity, reliability, and responsiveness of the EQ-HWB instruments across different informal caregiver contexts is now a priority to inform the instruments' final design.

Two measures, the EQ-HWB and EQ-HWB-S can be derived from the EQ-HWB instrument. The EQ-HWB consists of 25 items. The EQ-HWB-S is a short form with 9 items, which can be administered as its own instrument and may be more acceptable because of its shorter length.¹⁰ There is now a pilot UK value set for the EQ-HWB-S,¹¹ which has demonstrated that it can be valued using composite time trade-off with discrete choice experiment; further work is required to develop a full valuation protocol once the beta-version is released. Currently, there is no

Australian value set, and we have used the pilot UK value set as per our prior studies.^{12,13} In our previous validation work, we found that the EQ-HWB-S appeared to be valid and well accepted for caregivers of children whose families had experienced adversity; most convergent- and known-groups validity results conformed to a priori expectations, and we conducted in-depth interviews suggesting that the EQ-HWB-S was appropriate and well received.¹² In a second study, we compared the quality of life of parents of children with and without health conditions, finding moderate known-groups validity results and suggestions for potential item modifications.¹³ Construct validity of the EQ-HWB instruments for informal caregivers was supported in the United States.⁸ In an Australian general population sample comparing the new instruments with the EQ-5D, there were greater decrements in EQ-HWB-S utility values relative to the EQ-5D values for participants with mental health conditions.¹⁴

In our previous study of parents of children with health conditions,¹³ the sample was drawn from the pediatric multi-instrument comparison study (P-MIC).^{15,16} The sample size of 6257 participants allowed analysis by gender and child health, but there was limited parental information available for known-groups comparison, no comparative quality-of-life measures, and no test-retest information. Here, we used P-MIC as a sample frame to administer to conduct further analysis, including all 25 EQ-HWB items.

Our main aim was to provide an in-depth investigation of the validity and reliability of both EQ-HWB instruments in a sample of parents of children with health conditions. We assessed the response distributions of all 25 items, conducted convergent validity testing against relevant instruments, conducted known-group validity across a wide range of caregiver variables, and analyzed test-retest reliability. We were particularly interested in the instruments' ability to discriminate in terms of caregiving variables (time spent caregiving, financial impact, etc).

Methods

Study Design

Participants were drawn from the P-MIC sample frame (previous research has demonstrated the high quality of this sample^{13,16}). Of the participants included in P-MIC, 1050 participants, who were originally recruited through a Melbourne pediatric hospital, indicated that they would be interested in further research and could be approached for this study. From surveys sent to sample-frame participants, we received data from 225 caregivers at baseline, and 193 at 2-days follow-up (21% response rate at baseline from sample frame, 86% response rate at follow-up from baseline). Participants received an honorarium of AU\$25 after completing the second survey. Although there is no specific method to calculate sample size for psychometric validation studies for patient-reported outcome measures, 200 cases may potentially be suitable for testing reliability and validity.¹⁷ COSMIN guidelines note that samples of over 100 participants are "very good" for reliability analyses, construct validity, and comparing with other measurement instruments.¹⁸

Appendix Table 1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013> displays the 49 included child health conditions; many children had multiple conditions (57% had 2+). Conditions with the highest incidence were autism spectrum disorder (ASD, $n = 72$, 27.4%), attention-deficit/hyperactivity disorder ($n = 64$, 24.3%), developmental delay ($n = 48$, 18.6%), anxiety ($n = 44$, 16.7%), genetic conditions ($n = 38$, 14.5%), and asthma ($n = 32$, 12.2%). Ethics was obtained from the

relevant Human Research Ethics Committee (October 2023 Reference # 2023-27184-46174-5).

Procedure

The survey was informed by a desktop literature review of current best practice alongside discussions with experts on measuring caregiver attributes. A follow-up survey was sent 2 days after baseline. Surveys were accessed via a link embedded in an email. The baseline survey included demographic questions, EQ-5D-Y-5L proxy for the child, 3 caregiver quality-of-life instruments administered in random order (EQ-HWB, EQ-5D-5L, and CarerQol), and questions about caregiver burden and finances (see Appendix Box 1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013>). Demographic questions were asked about the caregiver and the study child, defined as the child with a medical condition or disability the caregiver was most concerned about. The follow-up survey included the EQ-HWB and several caregiver questions. Reminders were sent at days 3 and 4. Surveys were administered through RedCAP.¹⁹ We estimated the surveys would take less than 15 and less than 5 minutes, respectively to complete. We aimed to achieve a sample of 200 for the first survey.

Materials

Baseline characteristics for the study child included the following: length of study child's main or most significant medical condition (6 levels, see Table 1), gender (male/female/other), Aboriginal or Torres Strait Islander (yes/no), language other than English (yes/no), and child's mental health status as reported by parents (thriving/coping/struggling/always overwhelmed). Parent variables included gender (male/female/other/missing), highest education level (6 levels), current main activity (10 levels), household income (4 levels), household size, and parent holds an Australian Government Healthcare Card (indication of low income or receipt of government benefits; and provides a range of concessions including lower cost healthcare and pharmaceuticals) (yes/no).

The EQ-5D-Y-5L¹⁶ is the youth version of the EQ-5D and was included here in proxy form for the child (parents completed this questionnaire on behalf of the children). All 25 items of the experimental version of the EQ-HWB were included (which includes the 9 EQ-HWB-S items). The well-validated EQ-5D-5L was included to compare with the EQ-HWB,^{4,20} using the Australian value set.²¹ The CarerQol is a 7-item questionnaire developed specifically as a caregiver-related measure and thus a suitable comparator for the EQ-HWB because it includes mental health and quality-of-life questions.²² The CarerQol is validated in a range of Australian caregiver populations (including dementia, chronic diseases, children with special needs, and palliative care)²³ and has an Australian value set.²²

Caregiving burden included information on time spent caregiving, whether caregiving was shared, and other caregiving roles. Some items were adapted from the C-DEMQOL, a questionnaire specific to caregivers of people living with dementia,²⁴ and some from discussions with members of a Caregiver Consumer Advisory Group and lived-experience researchers. Questions in the financial impact section included whether the caregiver worked for pay, whether caregiving had affected the number of hours worked and promotion or job opportunities, and whether there were financial problems from caregiving. These items were derived from the C-DEMQOL and a study on the financial burden for families of children with congenital heart disease.²⁵ Three questions were asked regarding how well the caregiver felt supported (1) overall, (2) by professionals (for

Table 1. Baseline characteristics of parents of children living with a health condition ($n = 225$).

Parent variables (self-reported)	Group	n	%
Gender			
	Male	32	14.22
	Female	190	84.44
	Other	1	0.44
	Missing	2	0.89
Highest level of education			
	Bachelor degree or above	121	53.78
	Certificate III/IV, diploma or advanced diploma	65	28.89
	Certificate I/II	13	5.78
	Year 12	16	7.11
	Year 9-11	7	3.11
	Year 8 or below	1	0.44
	Missing	2	0.89
Current main activity (more than one option possible)			
	Full-time work	80	35.56
	Part-time work	66	29.33
	Casual work	29	12.89
	Unemployed and seeking work	8	3.56
	Full-time student	7	3.11
	Part-time student	15	6.67
	Full-time home duties	44	19.56
	Permanently retired	2	0.89
	On leave from work	8	3.56
	Other	11	4.89
Annual household income before tax*			
	\$25 999 or less per year	9	4.00
	\$26 000-\$51 999 per year	42	18.67
	\$52 000-\$103 999 per year	78	34.67
	\$104 000 or more per year	89	39.56
	Missing	7	3.11
Number of adults living in house (including parent)			
	1	31	13.78
	2	154	68.44
	3	21	9.33
	4	16	7.11
	5	1	0.44
	Missing	2	0.89
Number of children living in house (including study child)			
	1	66	29.33
	2	112	49.78
	3	31	13.78

Continued in the next column

Table 1. Continued

Parent variables (self-reported)	Group	n	%
	4	11	4.89
	5	1	0.44
	6	1	0.44
	7	1	0.44
	Missing	2	0.89
Holds a healthcare card (indicates low income or obtaining government benefits)			
	No	165	73.33
	Yes	58	25.78
	Missing	2	0.89
Study child variables (parent reported)			
	#	%	
Length of study child's main medical condition			
	Less than 1 month	3	1.33
	1-6 months	2	0.89
	7-12 months	8	3.56
	13-24 months	19	8.44
	More than 2 years	183	81.33
	Condition that changes	10	4.44
Gender			
	Male	137	60.89
	Female	84	37.33
	Other	4	1.78
Aboriginal or Torres Strait Islander			
	No	220	97.78
	Yes	5	2.22
Language other than English spoken at home			
	Yes	29	12.89
	No	196	87.11
Child's mental health over last 4 weeks			
	Thriving	63	28.00
	Coping	116	51.56
	Struggling	39	17.33
	Always overwhelmed	7	3.11

*Less than \$500 per week (\$25 999 or less per year); \$500-\$999 per week (\$26 000-\$51 999 per year); \$1000-\$1999 per week (\$52 000-\$103 999 per year); and \$2000 or more per week (\$104 000 or more per year) in Australian dollars.

example, doctors, social workers, and care workers), and (3) by family, friends, or community groups. We asked about the caregivers' general health and whether they had any chronic health conditions (yes/no). The caregiver general health question options (excellent/very good/good/fair/poor) were recoded to 2 groups (1 = excellent/very good/good, 2 = fair/poor) and 3 groups (1 = excellent/very good, 2 = good, 3 = fair/poor).

Statistical Analysis

We performed all analyses in STATA version 15. We calculated response distributions using frequency and percentages. The sum score was calculated by summing the scores of the individual

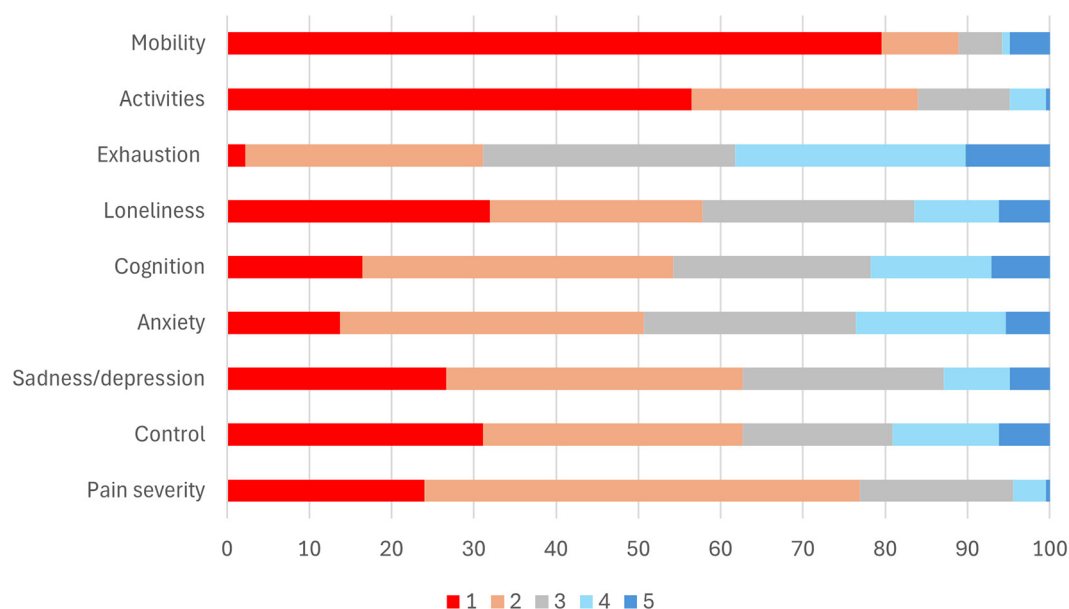
Table 2. Number and percentage of EQ-HWB item responses with means and standard deviations (SD) and Kappa values from test-retest results.

#	EQ-HWB item name	# (%) No difficulty	# (%) Slight difficulty	# (%) Some difficulty	# (%) A lot of difficulty	# (%) Unable	Mean	SD	Kappa
1	Seeing	111 (49.33)	69 (30.67)	31 (13.78)	4 (1.78)	10 (4.44)*	1.81	1.04	.437
2	Hearing	158 (70.22)	39 (17.33)	8 (3.56)	9 (4.00)	11 (4.89)*	1.56	1.07	.446
3	Mobility	179 (79.56)	21 (9.33)	12 (5.33)	2 (0.89)	11 (4.89)*	1.42	1.00	.343
4	Activities	127 (56.44)	62 (27.56)	25 (11.11)	10 (4.44)	1 (0.44)	1.65	0.88	.559
5	Personal care	181 (80.44)	24 (10.67)	16 (7.11)	4 (1.78)	0 (0)	1.30	0.68	.468
		None of the time	Only occasionally	Sometimes	Often	Most/all of the time			
6	Sleep problems	17 (7.56)	85 (37.78)	57 (25.33)	50 (22.22)	16 (7.11)	2.84	1.08	.573
7	Exhaustion	5 (2.22)	65 (28.89)	69 (30.67)	63 (28.00)	23 (10.22)	3.15	1.02	.534
8	Loneliness	72 (32.00)	58 (25.78)	58 (25.78)	23 (10.22)	14 (6.22)	2.33	1.20	.613
9	Felt unsupported	58 (25.78)	74 (32.89)	59 (26.22)	22 (9.78)	12 (5.33)	2.36	1.13	.500
10	Memory	56 (24.89)	76 (33.78)	52 (23.11)	31 (13.78)	10 (4.44)	2.39	1.13	.621
11	Cognition	37 (16.44)	85 (37.78)	54 (24.00)	33 (14.67)	16 (7.11)	2.58	1.14	.534
12	Anxiety	31 (13.78)	83 (36.89)	58 (25.78)	41 (18.22)	12 (5.33)	2.64	1.09	.528
13	Felt unsafe	167 (74.22)	31 (13.78)	20 (8.89)	7 (3.11)	0 (0)	1.41	0.78	.625
14	Felt frustrated	39 (17.33)	71 (31.56)	71 (31.56)	33 (14.67)	11 (4.89)	2.58	1.09	.531
15	Sadness/depression	60 (26.67)	81 (36.00)	55 (24.44)	18 (8.00)	11 (4.89)	2.28	1.09	.614
16	Nothing to look forward to	120 (53.33)	44 (19.56)	38 (16.89)	14 (6.22)	9 (4.00)	1.88	1.14	.688
17	Control	70 (31.11)	71 (31.56)	41 (18.22)	29 (12.89)	14 (6.22)	2.32	1.21	.610
18	Difficulty coping	111 (49.33)	54 (24.00)	36 (16.00)	18 (8.00)	6 (2.67)	1.91	1.10	.537
		Most/all the time	Often	Sometimes	Only occasionally	None of the time			
19	Felt accepted	64 (28.44)	81 (36.00)	39 (17.33)	34 (15.11)	7 (3.11)	2.28	1.13	.532
20	Felt good about self	29 (12.89)	65 (28.89)	77 (34.22)	45 (20.00)	9 (4.00)	2.73	1.05	.585
21	Could do things as wanted	30 (13.33)	60 (26.67)	68 (30.22)	55 (24.44)	12 (5.33)	2.82	1.11	.540
		None of the time	Only occasionally	Sometimes	Often	Most/all the time			
22	Pain frequency	53 (23.56)	80 (35.56)	60 (26.67)	26 (11.56)	6 (2.67)	2.34	1.05	.581
		No physical pain	Mild	Moderate	Severe	Very severe			
23	Pain severity	54 (24.00)	119 (52.89)	42 (18.67)	9 (4.00)	1 (0.44)	2.04	0.79	.659
		None of the time	Only occasionally	Sometimes	Often	Most/all the time			
24	Discomfort frequency	79 (35.11)	84 (37.33)	47 (20.89)	11 (4.89)	4 (1.78)	2.01	0.96	.495
		No physical discomfort	Mild	Moderate	Severe	Very severe			
25	Discomfort severity	70 (31.11)	123 (54.67)	27 (12.00)	5 (2.22)	0 (0)	1.85	0.71	.581

*Items including suspected incorrect (reversed) responses.

items (ie, adding each item score to a total, such that higher scores indicate lower quality of life), and thus unweighted. The index score was calculated by applying the preference-weighted scores from a recently published pilot UK value set,¹¹ noting the limitations we outlined in the introduction. We also reported time to complete each questionnaire. Analysis of the EQ-HWB was conducted at the sum-score level, and the EQ-HWB-S at the sum-score and index-score levels. We reverse scored the 3 positive EQ-HWB items (items 19–21) before analysis.

We assessed known-groups validity using student's *t* tests (2 groups) and one-way analysis of variances (3 groups) with hypothesized directions developed a priori. When comparing 2 groups, we used Cohen's *d* as the effect size, and for 3 groups we used eta squared. We hypothesized that in cases in which others shared in the care, the caregiver would have a higher quality of life because of reduced workload and responsibility. We hypothesized that caregivers would have lower quality of life if they had reduced their work hours or stopped working because of their

Figure 1. Distribution of EQ-HWB-S item responses at baseline.

caregiving responsibilities and if they had financial difficulties.²⁵ We hypothesized that in cases in which children were struggling/feeling overwhelmed, the caregiver would have poorer quality of life than if the child was thriving/coping²⁶ or if the child had lower scores on the EQ-5D-Y-5L index scores (lower quality of life). We hypothesized also that caregiving may be a risk factor for poor health and that this would be associated with poorer quality of life.²⁷ We hypothesized that in cases in which caregivers perceived that their needs for support were met (overall, by professionals, or by family/friends), they would have higher quality of life. We hypothesized that caregivers on lower incomes would have lower quality of life.²⁸ We note that caregiving burden has been found to be higher in women than men²⁹; however, our sample size was insufficient to examine gender differences.

Convergent validity was assessed using Spearman correlations comparing the EQ-HWB (25 items) against the EQ-5D-5L and the CarerQoL. A priori hypotheses were developed by first and final authors before analysis for items that expected to correlate with at least 0.3, a moderate correlation. Hypotheses were based on the EQ-5D being designed to focus on health-related aspects more narrowly and the CarerQoL on carer-related quality of life but the EQ-HWB aiming to capture both. The strength of the correlation was defined and interpreted as per Cohen 1992³⁰ (0.1-0.29 weak, 0.3-0.49 moderate, and ≥ 0.5 strong).

Surveys were sent 2 days apart to enable test-retest reliability. The time frame was chosen on the basis that there was little risk of changes in quality of life and participants are less likely to remember their previous response across the 2 days and following the recommendation of the QQuality OF Life in Kids: Key evidence to strengthen decisions in Australia (QUOKKA) protocol.³¹ Participants who had follow-up data were included ($n = 193$). To assess the degree of relatedness between the baseline and follow-up at the whole instrument level, we calculated intraclass correlation coefficients (ICC) using a mean rating, absolute-agreement, 2-way mixed-effects model for the EQ-HWB-S sum score as recommended by Koo and Li (2016).³² At the item level, we calculated Kappa scores.³³ An ICC of 0 to 0.5 was considered poor, 0.5 to 0.75 moderate, 0.75-0.9 good, and over 0.9 excellent reliability.³²

Weighted Kappa coefficients of 0 to 0.2 were considered poor, 0.21-0.40 fair, 0.41-0.6 moderate, 0.61-0.80 substantial, and 0.81 suggested close to perfect agreement.³³

Results

There were no missing data within in the surveys because participants were required to complete all items to proceed with the survey. Baseline characteristics from 225 participants who were parents of children with health conditions are presented in Table 1. Parents were an average age of 41.4 years ($SD = 7.4$ years). Children were aged 1 to 18, mean of 9.34 ($SD = 4.7$). A comparison of the baseline information with the original P-MIC sample can be found in Appendix Table 2 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013>³⁴

Response Distribution and Feasibility

The response distribution of the 25 EQ-HWB items is displayed in Table 2, including means (noting that the data are ordinal) and standard deviations (SD). We found negatively skewed distributions for the physical items (seeing, hearing, mobility, daily activities, personal care, and felt unsafe) with higher percentages in the lowest response levels. There were high levels of exhaustion, sleep problems, and anxiety, as well as 2 of the positive items, “felt good about self” and “could do things as wanted.” The distributions for the EQ-HWB-S are presented graphically in Figure 1. Time taken to complete questionnaires is included in Appendix Table 3 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013> in the supplemental files. Participants completed the EQ-HWB questions faster on average at follow-up (mean = 133 seconds, $SD = 135$) than at baseline (mean = 155 seconds, $SD = 176$).

There were higher than expected responses in the “unable” category for the first 3 EQ-HWB items, given that our previous work indicated that caregivers did not have high problems on these items.^{12,13} Comparison with the follow-up data suggests that these responses were likely to be incorrect. The distribution of the

Table 3. Effect sizes for known-group validity analysis for caregiver EQ-HWB sum scores, EQ-HWB-S index scores, EQ-HWB-S sum scores, EQ5D-5L index scores,²¹ and CarerQol index scores³⁵ using Cohen's *d* for student's *t* tests and eta squared for one-way analysis of variances.

Variables	EQ-HWB sum score	EQ-HWB-S sum score	EQ-HWB-S index score	EQ-5D-5L index score	CarerQol index score
Caregiver variables Cohen's <i>d</i> (2 groups)					
Others share in care	0.56	0.48	0.45	0.58	0.33
Caregiver contribution level	0.29	0.37	0.31	0.16	0.45
Work less due to child's condition	0.76	0.64	0.69	0.54	0.36
Stopped working	0.85	0.66	0.79	0.66	0.61
Having financial problems	1.29	1.04	1.20	0.84	0.87
Worried about meeting costs	1.16	0.94	1.01	0.63	0.94
Caregiver general health (high/low)	0.99	0.99	1.05	0.88	1.17
Carer has ongoing medical condition	0.54	0.65	0.55	0.27	0.49
Child variables					
Child's mental health	0.84	0.88	0.91	0.88	1.01
EQ-5D-Y	1.08	0.91	0.89	0.64	0.62
Caregiver variables Eta squared (3 groups)					
Caregiver needs for support met	0.39	0.30	0.29	0.12	0.09
Supported by professionals	0.34	0.22	0.20	0.09	0.07
Supported by family and community	0.37	0.29	0.23	0.05	0.09
Income	0.20	0.12	0.18	0.10	0.15
Caregiver general health (High/mid/low)	0.17	0.17	0.18	0.12	0.23
Child variables					
Child's mental health	0.22	0.19	0.20	0.13	0.18
EQ-5D-Y	0.34	0.28	0.26	0.18	0.16
Less than small effect size	0	0	0	1	0
Small effect size	1	2	2	1	3
Medium effect size	3	4	3	10	6
Large effect size	13	11	12	5	8

Note. Lightest blue indicates low effect size; medium blue, medium effect size; and darker blue, large effect size.
Cohen's *d*: 0.2-0.49 = small; 0.5-0.79 = moderate; and above 0.8 = large.
Eta squared: 0.01 = small; 0.06 = moderate; and 0.14 = large.

follow-up EQ-HWB items is displayed in Appendix Table 4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013>. When we compared baseline with follow-up data, it appeared that at baseline, 10 participants had incorrectly read the response options in the seeing variables and 11 in the Hearing and Mobility variables and that this response error was corrected when they got to item 4, "daily activities." We checked EQ-HWB item 3 (mobility) against EQ-5D-5L (mobility) and found that 11 people had indicated "unable" for EQ-HWB Mobility but "no problems" for EQ-5D-5L Mobility, suggesting that this anomaly was, indeed, a likely to be a response error (there were no other items to directly compare with the other 2 responses). This did not happen in the follow-up data. An investigation of the layout of the questionnaire did not indicate that the response options were difficult to read or understand. We have opted to retain these participant's responses but have highlighted the potential impact on results. Participants completed the EQ-5D-Y as a proxy report for their child just before completing the parent quality-of-life measures, and this may possibly have led to some confusion, although we clearly stated at the start of each questionnaire who was being reported for.

Known-Group Validity

We compared groups for a wide range of known groups across 5 instruments: EQ-HWB sum score, EQ-HWB-S index score, EQ-HWB-S sum score, EQ-5D-5L index score, and CareQol index score. Results are presented in Table 3^{21,35} with color coding for effect size strength; more detailed results are displayed in Appendix Tables 5a, 5b, 6a, and 6b in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013>. Results suggest that the EQ-HWB sum score, the EQ-HWB-S index score, and the EQ-HWB-S sum score performed well on the known-groups validity tests. The EQ-5D-5L performed less well on tests pertaining to caregiver contribution and support items, and the CarerQol was less able to detect differences between groups for the variable "whether others shared in the care" and for the support variables.

Convergent Validity

We had identified 9 correlations between the EQ-HWB and EQ-5D-5L items that we expected to correlate at least moderately (>0.3 and marked with §) and all a priori hypotheses were confirmed, as

Table 4. Spearman correlations between EQ-HWB and EQ-5D-5L items.

#	EQ-HWB items	EQ-5D-5L items					EQ-5D-5L index score	EQ-5D-5L sum score
		Mobility	Personal care	Usual activities	Pain/discomfort	Anxiety/depression		
1	Seeing	0.200*	-0.142 [†]	-0.156 [†]	-0.112	-0.033	0.162 [†]	-0.135 [†]
2	Hearing	-0.301 [†]	-0.275 [†]	-0.132	-0.196*	-0.051	0.173*	-0.167 [†]
3	Mobility	-0.485 ^{†,§}	-0.361 [†]	-0.259 [†]	-0.305 [†]	-0.095	0.303 [†]	-0.286 [†]
4	Activities	-0.424 [†]	-0.371 [†]	-0.458 ^{†,§}	-0.362 [†]	-0.294 [†]	0.437 [†]	-0.468 [†]
5	Personal care	-0.507 [†]	-0.603 ^{†,§}	-0.468 [†]	-0.387 [†]	-0.275 [†]	0.440 [†]	-0.489 [†]
6	Sleep problems	-0.118	-0.074	-0.251 [†]	-0.183*	-0.400 [†]	0.344 [†]	-0.356 [†]
7	Exhaustion	-0.154 [†]	-0.101	-0.245 [†]	-0.241 [†]	-0.414 [†]	0.355 [†]	-0.380 [†]
8	Loneliness	-0.162 [†]	-0.213*	-0.286 [†]	-0.180*	-0.531 [†]	0.365 [†]	-0.421 [†]
9	Felt unsupported	-0.241 [†]	-0.192*	-0.299 [†]	-0.230 [†]	-0.412 [†]	0.365 [†]	-0.409 [†]
10	Memory	-0.227 [†]	-0.174*	-0.274 [†]	-0.305 [†]	-0.415 [†]	0.370 [†]	-0.408 [†]
11	Cognition	-0.195*	-0.172*	-0.287 [†]	-0.255 [†]	-0.444 [†]	0.381 [†]	-0.416 [†]
12	Anxiety	-0.107	-0.168 [†]	-0.224 [†]	-0.209*	-0.607 ^{†,§}	0.381 [†]	-0.443 [†]
13	Felt unsafe	-0.166 [†]	-0.267 [†]	-0.205*	-0.035	-0.090	0.113	-0.139 [†]
14	Felt frustrated	0.016	-0.046	-0.206*	-0.089	-0.380 [†]	0.225 [†]	-0.271 [†]
15	Sadness/depression	-0.144 [†]	-0.224 [†]	-0.303 [†]	-0.214*	-0.634 ^{†,§}	0.425 [†]	-0.477 [†]
16	Nothing to look forward to	-0.249 [†]	-0.258 [†]	-0.357 [†]	-0.293 [†]	-0.494 [†]	0.435 [†]	-0.483 [†]
17	Control	-0.203*	-0.243 [†]	-0.344 [†]	-0.309 [†]	-0.547 [†]	0.448 [†]	-0.498 [†]
18	Difficulty coping	-0.199*	-0.256 [†]	-0.312 [†]	-0.249 [†]	-0.455 [†]	0.379 [†]	-0.427 [†]
19	Felt accepted	-0.372 [†]	-0.357 [†]	-0.410 [†]	-0.285 [†]	-0.415 [†]	0.429 [†]	-0.496 [†]
20	Felt good about self	-0.169 [†]	-0.200*	-0.306 [†]	-0.154 [†]	-0.366 [†]	0.297 [†]	-0.338 [†]
21	Could do things as wanted	-0.221 [†]	-0.145 [†]	-0.334 [†]	-0.219 [†]	-0.325 [†]	0.328 [†]	-0.373 [†]
22	Pain frequency	-0.402 [†]	-0.179*	-0.357 [†]	-0.637 ^{†,§}	-0.223 [†]	0.500 [†]	-0.506 [†]
23	Pain severity	-0.425 [†]	-0.264 [†]	-0.395 [†]	-0.642 ^{†,§}	-0.273 [†]	0.548 [†]	-0.543 [†]
24	Discomfort frequency	-0.400 [†]	-0.291 [†]	-0.344 [†]	-0.432 ^{†,§}	-0.239 [†]	0.448 [†]	-0.444 [†]
25	Discomfort severity	-0.364 [†]	-0.202*	-0.295 [†]	-0.546 ^{†,§}	-0.201*	0.461 [†]	-0.440 [†]
	EQ-HWB sum score	-0.366 [†]	-0.325 [†]	-0.451 [†]	-0.413 [†]	-0.544 [†]	0.555 [†]	-0.596 [†]
	EQ-HWB-S index score	0.336 [†]	0.299 [†]	0.436 [†]	0.423 [†]	0.575 [†]	-0.573 [†]	0.602 [†]
	EQ-HWB-S sum score	-0.314 [†]	-0.299 [†]	-0.419 [†]	-0.392 [†]	-0.626 [†]	0.565 [†]	-0.608 [†]

Note. White indicates $Rho < 0.3$; light blue, $0.3 < Rho < 0.5$; mid blue, $0.5 < Rho < 0.7$; and dark blue $Rho \geq 0.7$.

* $P < .01$.

[†] $P < .05$.

[‡] $P < .001$.

[§]Correlations hypothesized to be at least moderate (above 0.3).

shown in Table 4. We identified 31 correlations between the EQ-HWB and CarerQol items that we expected to be at least moderately correlated (>0.3 and marked with [§]), and 5 of the hypothesized correlations were less than moderate (in green), as shown in Table 5. We repeated the analysis comparing the EQ-5D-5L and the CarerQol to the EQ-HWB follow-up data, as shown in Appendix Tables 7 and 8 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.05.013>, which showed similar results.

Test-Retest Reliability

Intraclass correlation coefficients between baseline and 2 days follow-up for the 193 cases (86% of total sample) that we had test-retest data for suggested that reliability was good ($ICC = 0.85$) (95% CI 0.77-0.90, $F(192, 192) = 14.03$, $P < .001$) for the EQ-HWB, and good ($ICC = 0.83$) (95% CI 0.74-0.90, $F(192, 192) = 12.85$, $P < .001$) for the EQ-HWB-S raw scores.³⁶ For the individual item analyses,

Kappa results are displayed in the final column of Table 2. Kappa scores ranged from .343 (mobility) to .688 (nothing to look forward to), suggesting fair to substantial agreement,³³ and this was the same result for the EQ-HWB-S items. Kappa results were lowest for the first 3 items. Examination of these items revealed that, in the baseline data (as noted above in 3.1), 10 participants appeared to have reverse scored their responses in items 1 to 3. An examination of the presentation of the questions in the online survey revealed that there were no obvious reasons for this anomaly and these responses were retained in all analyses.

Discussion

We aimed to investigate the validity and reliability of the EQ-HWB and EQ-HWB-S in a sample of caregivers of children with a range of health conditions. We drew on a sample frame we had

Table 5. Spearman correlations between EQ-HWB and CarerQol items.

#	EQ-HWB items	CarerQol items							
		Fulfillment	Relational problems	Own mental health	Combining care tasks	Financial issues	Support	Own physical health	CarerQol index
1	Seeing	−0.098	0.229*	0.135 [†]	0.189 [†]	0.251*	−0.071	0.213 ^{‡,§}	−0.200 [‡]
2	Hearing	−0.137 [†]	0.108	0.093	0.100	0.169 [†]	−0.078	0.118 [§]	−0.089
3	Mobility	−0.160 [†]	0.183 [‡]	0.114	0.151 [†]	0.329*	−0.094	0.252 ^{‡,§}	−0.197 [‡]
4	Activities	−0.194 [‡]	0.224*	0.318*	0.387 ^{‡,§}	0.347*	−0.227*	0.432*	−0.352*
5	Personal care	−0.238*	0.155 [†]	0.178 [‡]	0.214 [†]	0.270*	−0.250*	0.237*	−0.135 [†]
6	Sleep problems	−0.157 [†]	0.221*	0.394*	0.387 ^{‡,§}	0.309*	−0.185 [‡]	0.438 ^{‡,§}	−0.407*
7	Exhaustion	−0.142 [†]	0.170 [†]	0.362*	0.414*	0.308*	−0.142 [†]	0.447 ^{‡,§}	−0.400*
8	Loneliness	−0.286*	0.256 ^{‡,§}	0.476*	0.421*	0.333*	−0.266 ^{‡,§}	0.394*	−0.336*
9	Felt unsupported	−0.292*	0.346*	0.391*	0.409*	0.378*	−0.339 ^{‡,§}	0.403*	−0.296*
10	Memory	−0.289*	0.161 [†]	0.387 ^{‡,§}	0.412*	0.348*	−0.182 [†]	0.382*	−0.316*
11	Cognition	−0.306*	0.265*	0.421 ^{‡,§}	0.402*	0.319*	−0.207 [‡]	0.367*	−0.332*
12	Anxiety	−0.173 [‡]	0.190 [‡]	0.520 ^{‡,§}	0.321*	0.282 ^{‡,§}	−0.160 [†]	0.304*	−0.360*
13	Felt unsafe	−0.263*	0.229 ^{‡,§}	0.203 [†]	0.144 [†]	0.239*	−0.159 [†]	0.182 [†]	−0.099
14	Felt frustrated	−0.156 [†]	0.357 ^{‡,§}	0.477*	0.514*	0.340*	−0.112	0.404*	−0.483*
15	Sadness/depression	−0.241*	0.246*	0.508 ^{‡,§}	0.322*	0.333*	−0.188 [‡]	0.402*	−0.360*
16	Nothing to look forward to	−0.313*	0.330*	0.440 ^{‡,§}	0.316*	0.364*	−0.223*	0.335*	−0.312*
17	Control	−0.222*	0.287*	0.502 ^{‡,§}	0.430*	0.445*	−0.224*	0.426*	−0.456*
18	Difficulty coping	−0.252*	0.329*	0.424 ^{‡,§}	0.340*	0.423*	−0.211 [†]	0.431*	−0.373*
19	Felt accepted	−0.334*	0.275*	0.332 ^{‡,§}	0.297*	0.311*	−0.344*	0.299*	−0.234*
20	Felt good about self	−0.375 ^{‡,§}	0.329*	0.462 ^{‡,§}	0.363 ^{‡,§}	0.391*	−0.261*	0.347*	−0.298*
21	Could do things as wanted	−0.347*	0.332*	0.448 ^{‡,§}	0.437 ^{‡,§}	0.474 ^{‡,§}	−0.255*	0.390*	−0.388*
22	Pain frequency	−0.130	0.157 [†]	0.198 [†]	0.165 [†]	0.303*	−0.075	0.445 ^{‡,§}	−0.285*
23	Pain severity	−0.114	0.139 [†]	0.212 [†]	0.134 [†]	0.313*	−0.114	0.418 ^{‡,§}	−0.267*
24	Discomfort frequency	−0.128	0.177 [†]	0.227*	0.262*	0.355*	0.004	0.394 ^{‡,§}	−0.303*
25	Discomfort severity	−0.041	0.140 [†]	0.131	0.208 [†]	0.282*	−0.098	0.370 ^{‡,§}	−0.253*
	EQ-HWB sum score	−0.349*	0.404*	0.559*	0.512*	0.549*	−0.299*	0.572*	−0.504*
	EQ-HWB-S index score	0.324*	−0.335*	−0.553*	−0.476*	−0.517*	0.296*	−0.553*	0.489*
	EQ-HWB-S sum score	−0.322*	0.329*	0.566*	0.501*	0.486*	−0.278*	0.545*	−0.488*

Note. White indicates $Rho < 0.3$; light blue, $0.3 < Rho < 0.5$; mid blue, $0.5 < Rho < 0.7$; and dark blue, $Rho \geq 0.7$; green are correlations were hypothesized to be over 0.3 but were not.

* $P < .001$.

[†] $P < .05$.

[‡] $P < .01$.

[§]Correlations hypothesized to be at least moderate (above 0.3).

previously used for an EQ-HWB-S validation study¹³ and expanded the scope in this study to include all 25 EQ-HWB items, a wide range of known groups, and 2 comparative quality-of-life instruments, the EQ-5D-5L and the CarerQol. We investigated item distribution, known-groups validity analysis, convergent validity analysis, and test-retest reliability.

Item distribution was good overall; however, the physical items (seeing, hearing, mobility, activities, personal care, and felt unsafe) were more likely to have negative distributions, with more than 50% of responses indicating no problem, compared with the other items, which were more normally distributed. Three items in the baseline data (personal care, felt unsafe, and discomfort severity) and 6 items in the follow-up data (hearing, activities, personal care, felt unsafe, pain severity, and discomfort severity)

had no responses in the highest response category, noting the likely data entry errors detailed in section “Response distribution and feasibility.” The distributions are similar to our previous articles in which we found that participants had high levels of exhaustion and anxiety but low levels of problems with mobility, daily activities, or pain^{12,13}

Known-group validity analysis suggested that the EQ-HWB sum score, the EQ-HWB-S index score, and the EQ-HWB-S sum scores were able to distinguish well between groups for all included analyses on a range of caregiver health and support variables and child mental health and child-health-related quality-of-life variables. These 3 versions of the EQ-HWB outperformed the EQ-5D-5L and the CarerQol in terms of effect sizes. The 3 EQ-HWB measures had an average of 11 results with

large effect sizes, compared with 5 for the EQ-5D-5L and 8 for the CarerQol. The EQ-HWB sum score slightly outperformed the EQ-HWB-S sum- and index-score measures, as could be expected given its greater number of items but not enough to warrant including a longer form when the short form has significantly less participant burden.¹⁰ However, this difference may not justify the addition of the extra 16 items unless researchers particularly want to include items that may be specific to caregivers such as support, coping, frustration, and felt unsafe. It was not possible to distinguish between the EQ-HWB-S sum- and index-score measures, based on these tests because some items had higher Cohen's *d* scores than others but with no clear pattern. We note, however, that the Cohen's *d* scores were higher in the EQ-HWB-S index score for both income and financial difficulties, than the EQ-HWB-S sum score. We note that this study presents far more detailed known-groups analysis than previous studies because it was set up with this particular aim and that these results support those in earlier articles with similar populations^{12,13}

The correlations between the EQ-HWB items and the EQ-5D-5L items conformed to expected hypotheses, and there were a further 38 correlations with at least moderate strength. The support item in the CarerQol had generally low correlations with the EQ-HWB items. It did not correlate at least moderately with EQ-HWB loneliness and was only just moderately correlated with the EQ-HWB support item. We note that the support item in the CarerQol is positively worded and comes after 4 negatively worded questions. It is therefore possible that a proportion of participants might therefore answer this question incorrectly in reference to the response options. We would have expected seeing, hearing, and mobility to correlate with the CarerQol item "own physical health," but they did not. The CarerQol item "relational problems" also did not correlate as expected with loneliness or felt unsafe. This is possibly because the CarerQol item addresses relationship problems specific to the care receiver, whereas the EQ-HWB items are not specific to that relationship.

One question that appears in the CarerQol but not the EQ-HWB is "financial problems because of my care tasks." We note that although the EQ-HWB does not specifically include a finance question, the known-group validity analysis did include a variable on financial problems, "Do you have any financial problems due to your child's condition? (yes/no)" taken from an article investigating financial burden in families of children with congenital heart disease.²⁵ The EQ-HWB measures had very high Cohen's *d* scores for this item, suggesting that they were well able to distinguish between groups in terms of financial burden. This is important evidence given the lack of a specific financial item in the EQ-HWB.

Strengths and Limitations

This work builds on our previous studies on the EQ-HWB-S that suggested that the EQ-HWB-S was a valid and reliable quality-of-life measure for caregivers of children under 8 years whose families had experienced adversity¹² and for parents of children with health conditions using a similar sample to this study.¹³ A main strength of this study was our extensive range of variables for known groups validity analysis which suggested that the EQ-HWB measures are valid in this population. Particular strengths were the inclusion of caregiving intensity and duration in the known-group analyses and a head-to-head comparison with the EQ-5D-5L and CarerQol. Test-retest results were similar to those in our previous study, which only had a small sample size,¹² and suggest that the instrument is reliable in this context.

Limitations of this study include that we administered the EQ-HWB 25-item instrument but did not separately administer the EQ-HWB-S; hence, these results need to be interpreted in this context. We did not have sufficient sample size to conduct a subgroup analysis by gender, as we had in our previous article.¹³ In terms of future analysis, longitudinal data would be of benefit in determining responsiveness to change over time, which is essential for determining the appropriateness of the EQ-HWB-S for producing QALYs for economic analysis. Such information would be important in the context of interventions or programs that improved the child's health condition or were aimed at lessening the caregiving burden. Future research may focus on gathering data over time and on other caregiver groups. There are limitations to using the pilot UK value set, which is the only one currently available. It will be important to investigate Australian-specific preference weights for the EQ-HWB-S as they become available. We also note the survey response rate and the possibility of differential nonresponse bias; the survey in the original P-MIC study from which this sample frame was drawn was very long and may have put people off who do not like completing surveys, thereby affecting the generalizability of the original sample, from which this sample frame was sourced. We were unable to compare our sample directly with the P-MIC sample as surveys were sent out by P-MIC researchers, and we did not have access to contact information to track them.

Conclusions

The EQ-HWB is a generic quality-of-life instrument that aims to measure health and well-being specific to a range of populations, including caregivers. We have demonstrated that both the long and short forms have good validity in the psychometric tests that we included in this study of parents of children with a range of health conditions. The wide range of known-groups specific to caregivers for validity analysis was a particular strength of this study, and both EQ-HWB measures performed well in determining differences between groups for caregiver duration and intensity. An important outcome from this study was that the EQ-HWB was able to distinguish important aspects of caregiving, although it is a generic scale. This study contributes to the emerging literature on the EQ-HWB instruments for use in measuring caregiver quality of life for economic analysis. This article provides further evidence to support one of the long-term goals of the "Extending the QALY" initiative, which is to provide the tools to include spillover effects in economic evaluations or the direct impacts of caregiving interventions.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section. Views expressed by the authors in the publication do not necessarily reflect the views of the EuroQol Foundation.

Supplemental Material

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.jval.2025.05.013>.

Article and Author Information

Accepted for Publication: May 22, 2025

Published Online: June 27, 2025

doi: <https://doi.org/10.1016/j.jval.2025.05.013>

Author Affiliations: Melbourne Health Economics, School of Population and Global Health, The University of Melbourne, Carlton, Victoria, Australia (Bailey, Dalziel); Murdoch Children's Research Institute, Melbourne, Victoria, Australia (Constable); Sheffield Centre for Health and Related Research, School of Medicine and Population Health, University of Sheffield, Sheffield, England, UK (Peasgood).

Correspondence: Cate Bailey, PhD, 207 Bouverie St, Carlton, 3035, Victoria, Australia. Email: cate.bailey@unimelb.edu.au

Authorship Confirmation: All authors certify that they meet the ICMJE criteria for authorship.

Funding/Support: This research was funded by the EuroQol Foundation (1559-RA).

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; or decision to submit the manuscript for publication.

Acknowledgment: The authors thank Alexandra Devine for her advice on the questions regarding caregiver support, finance, and work. The authors thank the members of the Caregiver Consumer Advisory Group, especially Kris Pierce, for their input to the authors understanding of issues related to caregivers of children with health conditions.

REFERENCES

1. EuroQol Research Foundation. EQ-HWB. Published online 2025 <https://euroqol.org/research-at-euroqol/eq-hwb/>. Accessed June 20, 2025.
2. Peasgood T, Mukuria C, Carlton J, Connell J, Brazier J. Criteria for item selection for a preference-based measure for use in economic evaluation. *Qual Life Res*. 2021;30(5):1425–1432.
3. Peasgood T, Mukuria C, Carlton J, et al. What is the best approach to adopt for identifying the domains for a new measure of health, social care and carer-related quality of life to measure quality-adjusted life years? Application to the development of the EQ-HWB? *Eur J Health Econ*. 2021;22(7):1067–1081.
4. EuroQol Research Foundation. EQ-5D. <https://euroqol.org/eq-5d-instruments/>; Published 2020. Accessed October 14, 2020.
5. Brazier J, Roberts J, Deverill M. The estimation of a preference-based measure of health from the SF-36. *J Health Econ*. 2002;21(2):271–292.
6. Brazier J, Peasgood T, Mukuria C, et al. *Extending the QALY (E-QALY) project: stages for developing a new generic measure*. Sheffield, England: The University of Sheffield; 2020.
7. Monteiro AL, Kuharic M, Pickard AS. A comparison of a preliminary version of the EQ-HWB short and the 5-level version EQ-5D. *Value Health*. 2022;25(4):534–543.
8. Kuharic M, Mulhern B, Sharp LK, Turpin RS, Pickard AS. Comparison of EQ-HWB and EQ-HWB-S with other preference-based measures among United States informal caregivers. *Value Health*. 2024;27(7):967–977.
9. EuroQol. EQ-HWB. <https://euroqol.org/information-and-support/euroqol-instruments/instruments-in-development/eq-hwb/>; Published 2024. Accessed August 5, 2024.
10. Norman R, Olsen JA. Expanding the scope of value for economic evaluation: the EQ-HWB. *Value Health*. 2022;25(4):480–481.
11. Mukuria C, Peasgood T, McDool E, Norman R, Rowen D, Brazier J. Valuing the EQ Health and wellbeing Short (EQ-HWB-S) using time trade-off and a discrete choice experiment: a feasibility study. *Value Health*. 2023;26(7):1073–1084.
12. Bailey C, Dalziel K, Constable L, et al. The performance of the EQ-HWB-S as a measure of quality-of-life of caregivers in families that have experienced adverse events. *Eur J Health Econ*. 2025;26(1):7–21.
13. Bailey C, Dalziel K, Jones R, et al. The validity of the EuroQol health and wellbeing short version (EQ-HWB-S) instrument in parents of children with and without health conditions. *Pharmacoeconomics*. 2024;42(Suppl 1):163–179.
14. Lee P, Engel L, Lubetkin E, Gao L. Exploring the comparability between EQ-5D and the EQ health and wellbeing in the general Australian population. *Value Health*. 2024;27(4):508–517.
15. QUOKKA Research Program. Quality OF Life in Kids: Key evidence to strengthen decisions in Australia. <https://www.quokkaresearchprogram.org/>; Published 2022. Accessed April 19, 2022.
16. Jones R, O'Loughlin R, Xiong X, et al. Comparative psychometric performance of common generic paediatric health-related quality of life instrument descriptive systems: results from the Australian paediatric multi-instrument comparison study. *Pharmacoeconomics*. 2024;42(suppl 1):39–55.
17. Frost MH, Reeve BB, Liepa AM, Stauffer JW, Hays RD. Mayo/FDA Patient-Reported Outcomes Consensus Meeting Group. What is sufficient evidence for the reliability and validity of patient-reported outcome measures? *Value Health*. 2007;10(suppl 2):S94–S105.
18. Morkink LB, Prinsen CA, Patrick DL, et al. COSMIN study design checklist for patient-reported outcome measurement instruments. https://www.cosmin.nl/wp-content/uploads/COSMIN-study-designing-checklist_final.pdf; Published 2019. Accessed January 23, 2024.
19. Harris PA, Taylor R, Minor BL, et al. The REDCap consortium: building an international community of software platform partners. *J Biomed Inform*. 2019;95:103208.
20. Long C, Mao Z, Yang Z. A head-to-head comparison of EQ-HWB and EQ-5D-5L in patients, carers, and general public in china. *Value Health*. 2024;27(7):848–856.
21. Norman R, Mulhern B, Lancsar E, et al. The use of a discrete choice experiment including both duration and dead for the development of an EQ-5D-5L value set for Australia. *Pharmacoeconomics*. 2023;41(4):427–438.
22. Hoefman RJ, van Exel J, Brouwer WB. Measuring the impact of caregiving on informal carers: a construct validation study of the CarerQol instrument. *Health Qual Life Outcomes*. 2013;11(1):1–13.
23. Voormolen DC, Van Exel J, Brouwer W, et al. A validation study of the CarerQol instrument in informal caregivers of people with dementia from eight European countries. *Qual Life Res*. 2021;30(2):577–588.
24. Brown A, Page TE, Daley S, et al. Measuring the quality of life of family carers of people with dementia: development and validation of C-DEMQL. *Qual Life Res*. 2019;28(8):2299–2310.
25. McClung N, Glidewell J, Farr SL. Financial burdens and mental health needs in families of children with congenital heart disease. *Congenit Heart Dis*. 2018;13(4):554–562.
26. Reed M, Bedard C, Perlman CM, Browne DT, Ferro MA. Family functioning and health-related quality of life in parents of children with mental illness. *J Child Fam Stud*. 2023;32(11):3366–3377.
27. Shiue I, Sand M. Quality of life in caregivers with and without chronic disease: Welsh Health Survey, 2013. *J Public Health*. 2017;39(1):34–44.
28. Guerra-Martín MD, Casado-Espinosa MDR, Gavira-López Y, Holgado-Castro C, López-Latorre I, Borrallo-Riego Á. Quality of life in caregivers of cancer patients: a literature review. *Int J Environ Res Public Health*. 2023;20(2):1570.
29. Rico-Blázquez M, Quesada-Cubo V, Polentinos-Castro E, et al. Health-related quality of life in caregivers of community-dwelling individuals with disabilities or chronic conditions. A gender-differentiated analysis in a cross-sectional study. *BMC Nurs*. 2022;21(1):69.
30. Cohen J. Quantitative methods in psychology: a power primer. *Psychol Bull*. 1992;112:1155–1159.
31. Jones R, Mulhern B, Devlin N, et al. Australian paediatric multi-instrument comparison (P-MIC) study: technical methods paper. <https://www.quokkaresearchprogram.org/project-1-1/>; Published 2023. Accessed June 25, 2025.
32. Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med*. 2016;15(2):155–163.
33. Landis J, Koch G. The measurement of observer agreement for categorical data. *Biometrics*. 1977;33(1):159–174.
34. Fair Work Commission. The national minimum wage. <https://www.fwc.gov.au/agreements-awards/minimum-wages-and-conditions/national-minimum-wage#:~:text=The%20national%20minimum%20wage%20sets,who%20are%20award%20free>. Accessed June 25, 2025.
35. Hoefman RJ, van Exel J, Brouwer WB. Measuring care-related quality of life of caregivers for use in economic evaluations: CarerQol tariffs for Australia, Germany, Sweden, UK, and US. *Pharmacoeconomics*. 2017;35(4):469–478.
36. Cicchetti DV. Guidelines, criteria, and rules of thumb for evaluating normed and standardized assessment instruments in psychology. *Psychol Assess*. 1994;6(4):284.